step by step

Betül Calisan - 03758752

PARAMETERS

height: Overall height of the cylinder and staircase
radius: Radius of the central cylinder
stair_width: Radial width of each stair tread
step_height: Target vertical rise between steps
steps_per_turn: Number of steps per 360° revolution
step_thickness: Thickness of each stair tread
door_width: Width of the door openings
door_height: Door height, limited by available clearance
door_interval: Number of steps between consecutive doors
clearance: Minimum vertical clearance between the doors and the staircase above (0.10 m)

RELATIONSHIPS

Cylinder height ↑ Number of steps ↑
More steps are generated while maintaining a consistent step height.
Steps per turn ↑ Angular spacing ↓
Steps are distributed more closely around the cylinder.
Stair position → Door position
Each door follows the rotation and elevation of its corresponding step.
Stair spacing → Door height
Door height is limited by the vertical clearance between staircase revolutions.
Door geometry → Boolean Difference
Cubes are subtracted from the cylinder to create openings.
Surface classification → Colour
Exterior surfaces are blue, door opening surfaces are yellow, and stairs are red, for to dedicate the materials for rendering easier

